

Samet Kocbay

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M.Sc. student in Computational Science and Engineering focused on computational physics, PDE-based modeling, and scientific machine learning

PROFESSIONAL EXPERIENCE

- **Max Planck Institute for Plasma Physics**  October 2025 - September 2026
Working Student / Thesis
 - Derived and implemented a no-wall boundary condition extension for the nonlinear MHD code JOEREK
 - Extended validated tokamak cases to stellarator configurations via GVEC equilibria to verify the novel coupling
- **BMW Group**  October 2024 - September 2025
Working Student
 - Built surrogate models for operator-valued mapping from structural parameters to frequency responses
 - Improved prediction accuracy by >40% using multi-task Gaussian Processes
 - Reduced computational cost by 80% via latent representation learning
- **Compact Dynamics**  March 2024 - September 2024
Bachelor Thesis
 - Developed an RNN-based thermal neural network for rotor temperature estimation in electric motors, achieving 3–6 °C prediction error
 - Deployed the model for real-time inference on an embedded control unit (ECU)
 - Resolved the cold-start initialization problem by using back-EMF as a physics-based correction to the recurrent state
- **KIT - Institute of Fluid Mechanics**  July 2023 – December 2023
Student Assistant
 - Reconstructed velocity/pressure fields from sparse PIV measurements using PINNs, enabling reduced experimental measurement requirements

EDUCATION

- **M.Sc. Computational Science and Engineering** October 2024 - November 2026
Technical University of Munich (TUM), Current Grade: 1.5 (German system)
- **B.Sc. Mechanical Engineering** October 2019 - September 2024
Karlsruhe Institute of Technology (KIT)

PROJECTS

- **KA RaceIng: Head of Aerodynamics** October 2021 - September 2023
 - Led 13-person team, over a full aerodynamic development cycle (CFD → manufacturing)
 - Designed and optimized aero package (+14% downforce) using RANS-based CFD (k- ω SST)
 - Improved on mesh generation, boundary layer resolution (y+ control) and conducted grid convergence studies
- **High Performance Computing — Barnes–Hut** 
 - Built a Barnes–Hut solver from scratch in C++, parallelised with two different strategies:
 - * **OpenMP & AVX-512** — task-based parallel octree construction and vectorised force kernels
 - * **MPI** — Morton (Z-order) curve domain decomposition, Local Essential Trees

SKILLS

- **Programming:** Python, Fortran, Linux, Bash, C++, MATLAB, OpenMP, MPI
- **Machine Learning & SciML:** PyTorch, GPyTorch, PINNs, Neural Operators, Gaussian Processes
- **CAE & CFD:** STAR-CCM+, OpenFOAM (RANS, k- ω SST), Composite Optimization, FEM, FVM
- **Languages:** German (Native), English (Fluent), Turkish